

Analysis III

Home Work 2

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Exercise 1 (Step functions, 1+2+1 P.)

Let us define, for $x \in [0; 1]$,

$$f(x) = \sum_{k=0}^{+\infty} \chi_{[2^{-k-1}, 2^{-k}]}(x).$$

(a) Is f a step function? Justify your answer briefly (you do not need to give a complete proof).

(b) Show that there exists a step function $g : [0; 1] \rightarrow \mathbb{R}$ with $f = g$ almost everywhere and explicitly write down the set

$$\{x \in [0; 1] : f(x) \neq g(x)\}.$$

(c) Compute $\int_0^1 f(x) dx$ by making use of (b).

Answer:

(a) $f(x)$ is NOT a valid step function. A valid step function can always be represented by a finite number of bounded intervals. The sum shown is not finite and can't be represented as a finite number of bounded step functions (there are infinitely many isolated points that take the value 2).

(b) The simple step function: $g : [0, 1] \rightarrow \mathbb{R}$ given by: $g(x) = \chi_{[0,1]}(x)$ is such that $f = g$ (a.e.).

Now the ends of the each interval: $\chi_{[2^{-k-1}, 2^{-k}]}$ are "double counted" - except for $k = 0$ when $\chi_{[2^{-k-1}, 2^{-k}]} = \chi_{[\frac{1}{2}, 1]}$.

Consequently, the points where $f \neq g$ is the set:

$$D = \left\{ \frac{1}{2^k} \mid k \in \mathbb{N}_+ \right\} \cup \{0\}$$

Not that D is countable and hence a set of measure zero. I.e. $f = g$ (a.e.).

(c) It is clear that g is Lebesgue Integrable.

Using Theorem 11.2.15 in the notes we can conclude that, for Lebesgue Integration:

$$\int g = \int f$$

As $\int g = \lambda([0, 1]) = 1$, we can conclude that:

$$\int f = 1$$

Exercise 2 (Sequences of step functions, 2+1+2+1 P.)

Let I be a compact interval in $\mathbb{R}^n$.

- (a) Let $f : I^\circ \rightarrow \mathbb{R}$ be continuous and defined on the interior I° of I . Assuming that f is unbounded from below, show that there cannot exist an increasing sequence $(f_k)_{k \in \mathbb{N}} \in T^{\text{inc}}$ of step functions with $f_k \rightarrow f$ almost everywhere.
- (b) Let $(f_k)_{k \in \mathbb{N}} \in T^{\text{dec}}$ be a decreasing sequence of step functions for which $f_k \rightarrow \chi_I$ almost everywhere. Show that

$$\int_I f_k(x) dx \rightarrow \lambda_n(I)$$

by reducing the problem to an application of Theorem 11.2.10(1).

- (c) Consider the sequence of step functions $f_k : [0; 1] \rightarrow \mathbb{R}$ given by

$$f_k(x) = k\chi_{[0; 1/k]}(x).$$

Prove that $f_k \rightarrow 0$ pointwise almost everywhere and compute $\int_0^1 f_k(x) dx$ for every $k \in \mathbb{N}$. Why is this not in contradiction with Theorem 11.2.10(1)?

- (d) Let $f, g : I \rightarrow \mathbb{R}$ be two functions so that $f = g$ almost everywhere. Show that if $f \in \mathcal{L}^{\text{inc}}$, then also $g \in \mathcal{L}^{\text{inc}}$ and

$$\int_I f(x) dx = \int_I g(x) dx.$$

Answer:

- (a) Suppose such an increasing sequence (f_k) exists. Since f_1 is a step function, there is $K \in \mathbb{R}$ such that $f_1(x) \geq K$ everywhere. Thus

$$f_k(x) \geq f_1(x) \geq K.$$

Taking limits gives $f(x) \geq K$ almost everywhere.

Since f is unbounded below, choose $x_0 \in I^\circ$ with $f(x_0) < K$. By continuity, there exists $r > 0$ such that $B(x_0, r) \subset I^\circ$ and

$$f(x) < K \quad \text{for every } x \in B(x_0, r).$$

This ball has positive measure, contradicting $f \geq K$ almost everywhere. Therefore no such sequence exists.

- (b) Define the step functions

$$h_k = (f_k - \chi_I)\chi_I.$$

Then (h_k) is decreasing and $h_k \rightarrow 0$ almost everywhere. By Theorem 11.2.10(1),

$$\int h_k(x) dx \longrightarrow 0.$$

Therefore

$$\begin{aligned} \int_I f_k(x) dx &= \int h_k(x) dx + \int_I 1 dx \\ &= \int h_k(x) dx + \lambda_n(I) \longrightarrow \lambda_n(I). \end{aligned}$$

(c) Fix $x \in (0, 1]$. Choose $K > 1/x$. For every $k \geq K$, $1/k < x$, so $f_k(x) = 0$. Thus $f_k(x) \rightarrow 0$ for every $x > 0$. At $x = 0$, $f_k(0) = k \rightarrow +\infty$. Since $\{0\}$ is null, convergence holds almost everywhere.

For every $k \geq 1$,

$$\int_0^1 f_k(x) dx = k \lambda([0, 1/k]) = k \frac{1}{k} = 1.$$

There is no contradiction with Theorem 11.2.10(1): the sequence is not decreasing. Indeed,

$$f_{k+1}(x) = k + 1 > k = f_k(x) \quad \text{for } 0 < x < \frac{1}{k+1}.$$

Exercise 3 - Non-integrable function

Consider

$$f(x) = \frac{1}{x}$$

for $x \in (0; 1]$.

- (a) Construct an increasing sequence $(f_k)_{k \in \mathbb{N}} \in T^{\text{inc}}$ of step functions on $(0; 1]$ so that $f_k \rightarrow f$ almost everywhere.
- (b) Let $(g_k)_{k \in \mathbb{N}} \in T^{\text{inc}}$ be any increasing sequence of step functions with $g_k \rightarrow f$ almost everywhere. Show that the sequence

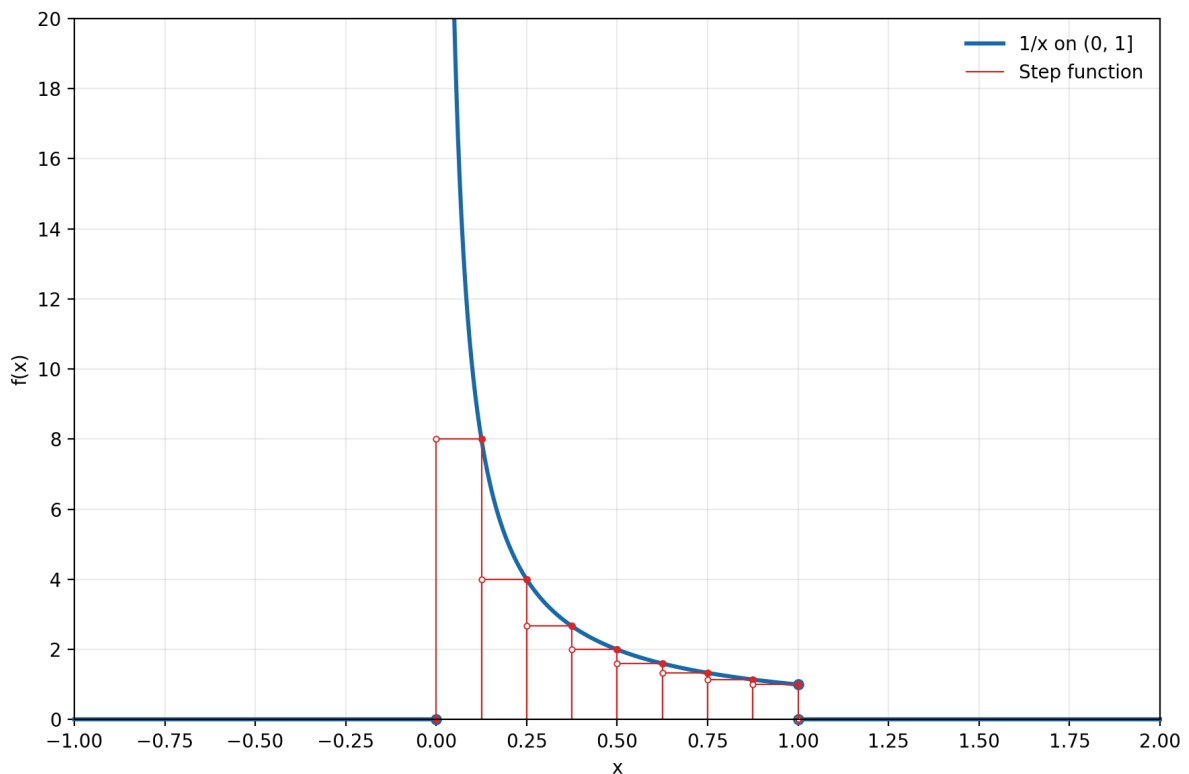
$$\left(\int_0^1 g_k(x) dx \right)_{k \in \mathbb{N}} \quad (3.1)$$

is unbounded. Is $f \in \mathcal{L}^{\text{inc}}$?

Answer:

(3)(a) This idea is copied from our lecture.

I have show a graph of the region (approx) we are interested in (with $x \in (0, 1]$).



The red rectangles are the step function:

$$f_k(x) = \begin{cases} 0, & x \leq 0, \\ \frac{N_k}{j}, & x \in \left(\frac{j-1}{N_k}, \frac{j}{N_k} \right], \quad j = 1, \dots, N_k, \\ 0, & x > 1, \end{cases}$$

where $N_k = 2^k$.

In the graph I have plotted $f_3(x)$. Monotonicity is obvious and $(f_k)_{k \in \mathbb{N}} \in T^{inc}((0, 1], \mathbb{R}_{>0})$. Let's prove pointwise convergence:

Fix $x \in (0, 1]$. Let b_k be the right endpoint of the interval containing x . Then

$$0 \leq b_k - x < 2^{-k}, \quad f_k(x) = \frac{1}{b_k}.$$

Therefore

$$0 \leq \frac{1}{x} - f_k(x) = \frac{b_k - x}{xb_k} \leq \frac{2^{-k}}{x^2} \rightarrow 0.$$

Hence $f_k(x) \rightarrow 1/x$.

(b) We have proved (in lectures) the following theorem:

Theorem 11.2.10 (1) Let $(f_k)_{k \in \mathbb{N}} \subset T$ be a monotonically decreasing sequence of step functions with $f_k \xrightarrow{k \rightarrow \infty} 0$ a.e. in $\mathbb{R}$. Then:

$$\lim_{k \rightarrow \infty} \int f_k = 0.$$

We shall show that the limit of this Lebesgue integral does NOT exist for our $(f_k)_k$ from part (a).

For each of the $N_k = 2^k$ rectangles, the area is:

$$\frac{1}{2^k} \frac{2^k}{j} = \frac{1}{j}.$$

Hence the total area is:

$$A_k = \sum_{j=1}^{2^k} \frac{1}{j}.$$

This is a well known series and diverges as $k \rightarrow +\infty$.

Now let $(g_k)_{k \in \mathbb{N}} \in T^{inc}$ be any increasing sequence of step functions with $g_k \rightarrow f$ almost everywhere.

Fix $A > 0$. Since $A_m \rightarrow +\infty$, choose m such that

$$\int_0^1 f_m(x) dx = A_m > A + 1.$$

Keep m fixed and define

$$h_k = \max(f_m - g_k, 0).$$

Each h_k is a step function. Since g_k increases to f almost everywhere and $f_m \leq f$, h_k decreases to 0 almost everywhere. By Theorem 11.2.10(1),

$$\int_0^1 h_k(x) dx \longrightarrow 0.$$

Thus there exists K such that, for every $k \geq K$,

$$\int_0^1 h_k(x) dx < 1.$$

Since $f_m \leq g_k + h_k$, for every $k \geq K$,

$$\begin{aligned} \int_0^1 g_k(x) dx &\geq \int_0^1 f_m(x) dx - \int_0^1 h_k(x) dx \\ &> A. \end{aligned}$$

Since $A > 0$ was arbitrary,

$$\int_0^1 g_k(x) dx \longrightarrow +\infty.$$

Therefore the sequence is unbounded, and $f \notin \mathcal{L}^{\text{inc}}$.